QPCode **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) BSC DOUBLE MAJOR EXAMINATION OCTOBER 2025

THIRD SEMESTER

DSC B - MG3DSCECC202 - NETWORKING FUNDAMENTALS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1.	List two applications of networks.	[K] / [CO1]
2.	Compare digital and analog signals in terms of accuracy.	[An] / [CO1]
3.	List two factors that affect bandwidth.	[U] / [CO1]
4.	Define a standard.	[K] / [CO1]
5.	Which spread spectrum technique is commonly used in Wi-Fi?	[A] / [CO2]
6.	Name any one function of the Data Link Layer.	[K] / [CO3]
7.	Define a noiseless channel.	[K] / [CO3]
8.	Identify the IEEE standard for 10 Gigabit Ethernet over twisted pair cables.	[K] / [CO3]
9.	State any two advantages of using Bluetooth for wireless communication.	[K] / [CO3]
10.	Define firewall in computer networks?	[K] / [CO4]

[2x10 = 20]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

11.	Compare attenuation, distortion, and noise.	[An] / [CO1]
12.	Explain how FDM (Frequency Division Multiplexing) works.	[U] / [CO2]
13.	In which scenario would a datagram network be more efficient than a virtual-circuit network?	[A] / [CO2]
14.	Differentiate between noiseless and noisy channels.	[U] / [CO3]
15.	Compare the functioning of a bridge and a router in terms of data handling and efficiency.	[U] / [CO4]
16.	Distinguish between static IP addressing and dynamic IP addressing in both IPv4 and IPv6.	[U] / [CO4]
17.	Define the three-way handshake in TCP.	[K] / [CO4]

18.	Name and briefly describe the function of at least three common application layer protocols	[K] / [CO4]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

19.	Discuss in detail the seven layers of the OSI model with neat diagram.	[U] / [CO1]
20.	In an emergency communication system, which type of transmission medium would you prefer, guided or unguided media? Explain why.	[An] / [CO2]
21.	Differentiate between Pure ALOHA, Slotted ALOHA, and CSMA in terms of efficiency, delay, and collision handling.	[U] / [CO3]

[10x2 = 20]